

MEDICAL STUDY WEBAPP DESIGN SYSTEM

Product feel

Premium, immersive, academic, calm, modern, information-rich, comfortable for long sessions.

Avoid generic dashboards, raw Markdown styling, marketing landing-page aesthetics, and toy-like UI.

Desktop

Preferred structure:

- right-side navigation
- central reading area
- optional contextual controls

Use efficient wide-screen space without making text lines excessively long.

Tablet

Reduce side chrome intelligently. Keep navigation and touch targets comfortable.

Mobile

Use a navigation drawer/sheet or compact control. Keep reading surfaces full width. Avoid controls covering content.

The page itself must never scroll horizontally, at any width. See "Zero horizontal overflow" below.

Zero horizontal overflow

This applies at every breakpoint, especially phone widths (320–430px):

- Every container uses `box-sizing: border-box`.
- No fixed-pixel-width element may exceed the viewport; prefer `%`, `vw` (capped), `min()` / `max()` / `clamp()`.
- Tables get their own scrollable wrapper (`overflow-x: auto` on the wrapper, not the page).
- Long strings (URLs, drug names, mixed-language terms) wrap instead of stretching their container.
- Decorative/animated background layers are sized and clipped so they cannot expand the scrollable area.
- `overflow-x: hidden` on the root is a safety net only — the underlying cause of any overflow must actually be fixed, not just hidden.

Right-side navigation

Hierarchy:

Chapter

Heading

Subheading

Each chapter expands/collapses independently.

Use logical indentation and active-state highlighting.

Mobile navigation drawer

- Width constrained to the viewport (e.g. ``min(85vw, <max>)``) — never a fixed width that can exceed a narrow phone screen.
- Because the app is RTL, the drawer's open/close slide direction must be verified explicitly for RTL — it must slide in from the correct logical edge, not assume an LTR default.
- The drawer must be fully visible within the viewport in its open state; no portion should render off-screen.
- Internal scroll for long chapter lists; the page behind it must not grow or scroll horizontally while the drawer is open.
- Clear close control plus tap-outside-to-close, both reachable at every breakpoint.

RTL

Use true directionality, not just alignment.

Use CSS logical properties and direction-aware icons.

Never use fixed LTR assumptions for:

- chevrons
- previous/next
- breadcrumbs
- nested indentation
- directional diagrams

- drawer slide direction

Mixed Persian/English medical content needs bidi-safe handling.

Themes

Provide true Light and Dark themes using semantic design tokens.

Tokens should cover:

background, surface, elevated surface, primary text, secondary text, border, accent, success, warning, danger.

Dark theme must be as rich as light theme

A dark theme that reads as "flat gray-on-black" is a failure state, not an acceptable minimal dark mode. Requirements:

- At least three distinguishable dark surface levels (base, surface, elevated) for visible depth.
- Ambient gradients/animated backgrounds stay visually present in dark mode — use deep, saturated hues (deep blue, violet, muted teal, warm dark accent) rather than desaturating everything to gray.
- Accent colors used for chapter markers, active states, and interactive elements keep their hue character in dark mode, adjusted only for contrast — they should not collapse into the same one or two neutral grays used for structure.
- Borders/separators are visible without being harsh.
- Primary text is a soft off-white rather than pure white, for comfortable long-session reading; secondary text keeps clearly lower but still legible contrast.

Animated backgrounds

Use subtle animated gradient meshes, blurred orbs, ambient glows, or slow geometric movement.

Keep them behind readable surfaces and low contrast in both themes — reduced but not eliminated in dark mode (see Dark theme section).

Respect reduced motion.

Ensure these layers are sized/clipped so they can never cause horizontal overflow.

Gradients

Use rich but coordinated gradients in:

- chapter headers
- title areas
- navigation accents
- section separators
- subtle surfaces
- interactive states

Avoid random rainbow styling. Keep gradients equally intentional in dark mode, not muted into near-invisibility.

Fonts

User-selectable font family:

- Vazirmatn
- Shabnam
- Sahel

Font size: continuous zoom, not discrete tiers

Do not offer named size tiers (Compact/Default/Large). Instead provide an increase (+) and decrease (-) control:

- Repeated presses step the base font size up/down by a small consistent increment.
- Enforce a minimum and maximum so layout never breaks.
- Size is driven by a root-level relative unit (CSS custom property / rem) so headings, body, tables, and navigation scale together.
- Give lightweight visual feedback of the current level.
- Persist both font-family and font-size preferences locally where practical.

Reading ergonomics

Prioritize readable line length, strong hierarchy, adequate line height, clear tables, restrained emphasis, and comfortable spacing at every font-size step within the enforced range.

Visual variation

Use meaningful variety:

- callouts
- comparison tables
- algorithms
- step blocks
- section accents
- expandable details
- navigation landmarks

Do not turn every paragraph into a card.

Motion

Use subtle transitions for interaction, expansion, navigation, and theme changes.

No distracting perpetual motion.

Long-document UX

For large libraries:

- hierarchical navigation
- search
- stable scroll position where practical
- direct anchors
- bookmarks/progress where useful
- compact navigation on mobile

Accessibility

Semantic headings, keyboard support, visible focus, accessible labels, contrast, touch-friendly controls, reduced motion, no color-only semantics.